

Software Development

AGILE object oriented software design and development

IDENTIFICATION

CODE : IF-4-PLD-AGILE
ECTS : 4.0

HOURS

Lectures : 13.5 h
Seminars : 0.0 h
Laboratory : 32.0 h
Project : 0.0 h
Teacher-student
contact : 45.5 h
Personal work : 50.0 h
Total : 95.5 h

ASSESSMENT METHOD

Evaluation of the project: Written report (by groups of 6 students), and oral presentations (at mid-time and at the end of the projects)

TEACHING AIDS

Slides
Java source code of an application [PlaCo] used to illustrate design patterns

TEACHING LANGUAGE

French

CONTACT

MME SOLNON Christine
christine.solnon@insa-lyon.fr

AIMS

A first goal is to introduce how to design object oriented softwares (structuration by means of packages and classes that are weakly coupled and strongly cohesive; utilisation of design patterns).

A second goal is to study iterative Agile methodologies for designing, implementing and maintaining object oriented software.

A third goal is to study how to adopt a quality process within an Agile context.

Main targeted skills:

- Apply Agile iterative methodologies to design, implement and maintain softwares
- Use UML diagrams to model
- Design the object oriented architecture of a software
- Apply object oriented principles to develop softwares
- Choose suited algorithms and data structures
- Apply generic tools for software quality

CONTENT

- 1 - Principles of Object Oriented Design and Design Patterns
- 2 - Agile and Iterative Software Development
- 3 - Tools for Software Quality

Practice of an Agile methodology to analyze, design, develop and test a software. This project is also an opportunity to use design patterns, tools for software quality control, test-driven development, versioning, continuous integration, model-checking, etc.

BIBLIOGRAPHY

UML 2 et les design patterns. Craig Larman
Tête la première : Design Patterns. Eric Freeman & Elizabeth Freeman
Meyer, Bertrand. "Touch of class." Learning to program well with Object Technology and Design by Contract
CRISPIN, GREGORY, Agile Testing, 2009
WHITTAKER, ARBON, CAROLLO, how google tests software, 2012
MYERS, the art of software testing, 2012
RILEY, beautiful testing, 2010

PRE-REQUISITE

IF-3-GL
IF-3-ALGO
IF-3-POO
IF-3-POOA
IF-3-AAIA

INSA LYON

Campus LyonTech La Doua

20, avenue Albert Einstein - 69621 Villeurbanne cedex - France
Phone +33 (0)4 72 43 83 83 - Fax +33 (0)4 72 43 85 00
www.insa-lyon.fr